

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$\frac{1}{3}(x + 6) - \frac{1}{2}(x + 6) = -8$$

What value of x is the solution to the given equation?

Correct Answer: 42

Rationale

The correct answer is **42**. The expression $(x + 6)$ is a factor of both terms on the left-hand side of the given equation. Therefore, the given equation can be written as $(x + 6)\left(\frac{1}{3} - \frac{1}{2}\right) = -8$, or $(x + 6)\left(-\frac{1}{6}\right) = -8$. Multiplying each side of this equation by -6 yields $x + 6 = 48$. Subtracting **6** from each side of this equation yields $x = 42$. Therefore, the value of x that is the solution to the given equation is **42**.